

Investment Research Assistant

Fine-Tuned GPT Model + Streamlit Application

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DSC 670 Advanced Uses of Generative AI

Milestone 4

Winter 2025-2026

Project Overview

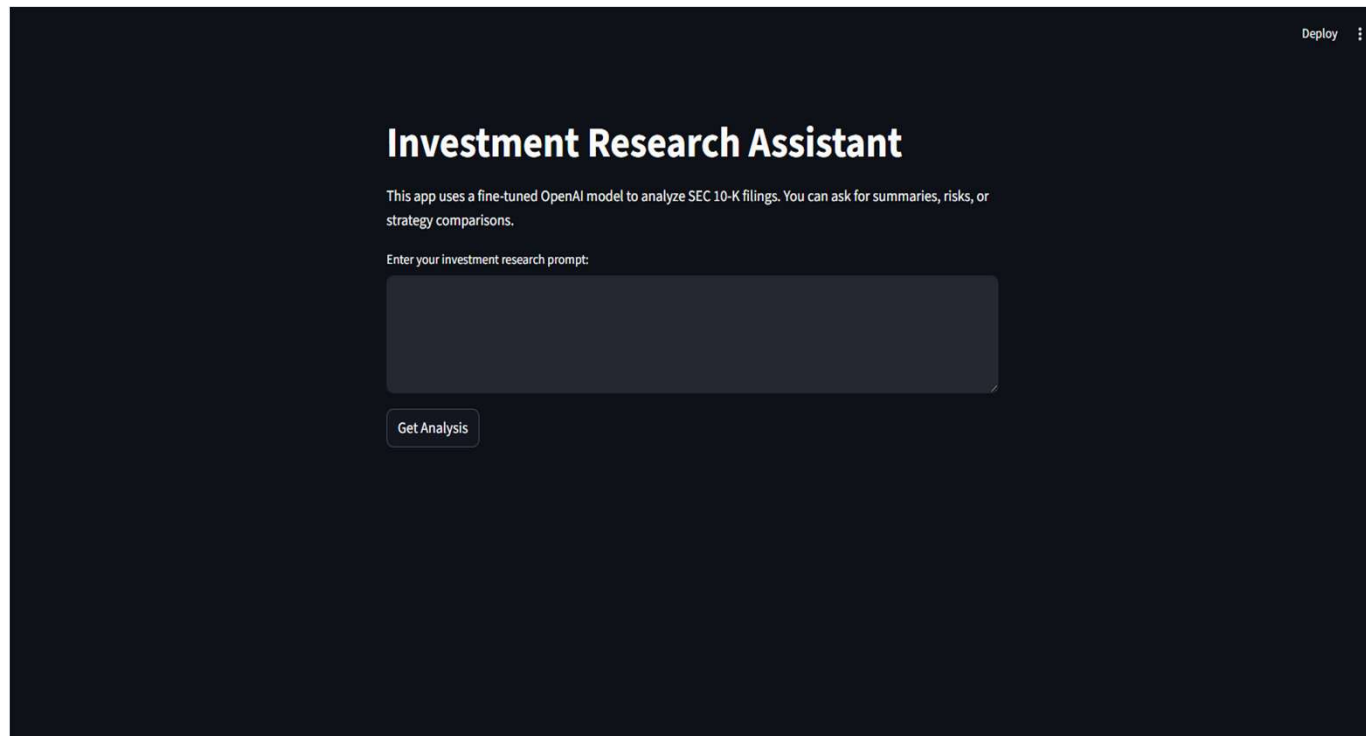
- Developed a fine-tuned GPT-3.5 model
- Focused on SEC 10-K style financial analysis
- Designed to generate analyst-style summaries and risk assessments
- Built an interactive web application using Streamlit

Model Fine-Tuning

- Base Model: gpt-3.5-turbo
- Custom training data for investment research prompts
- Structured responses for summaries and risk analysis
- Goal: Improve consistency and financial reasoning quality

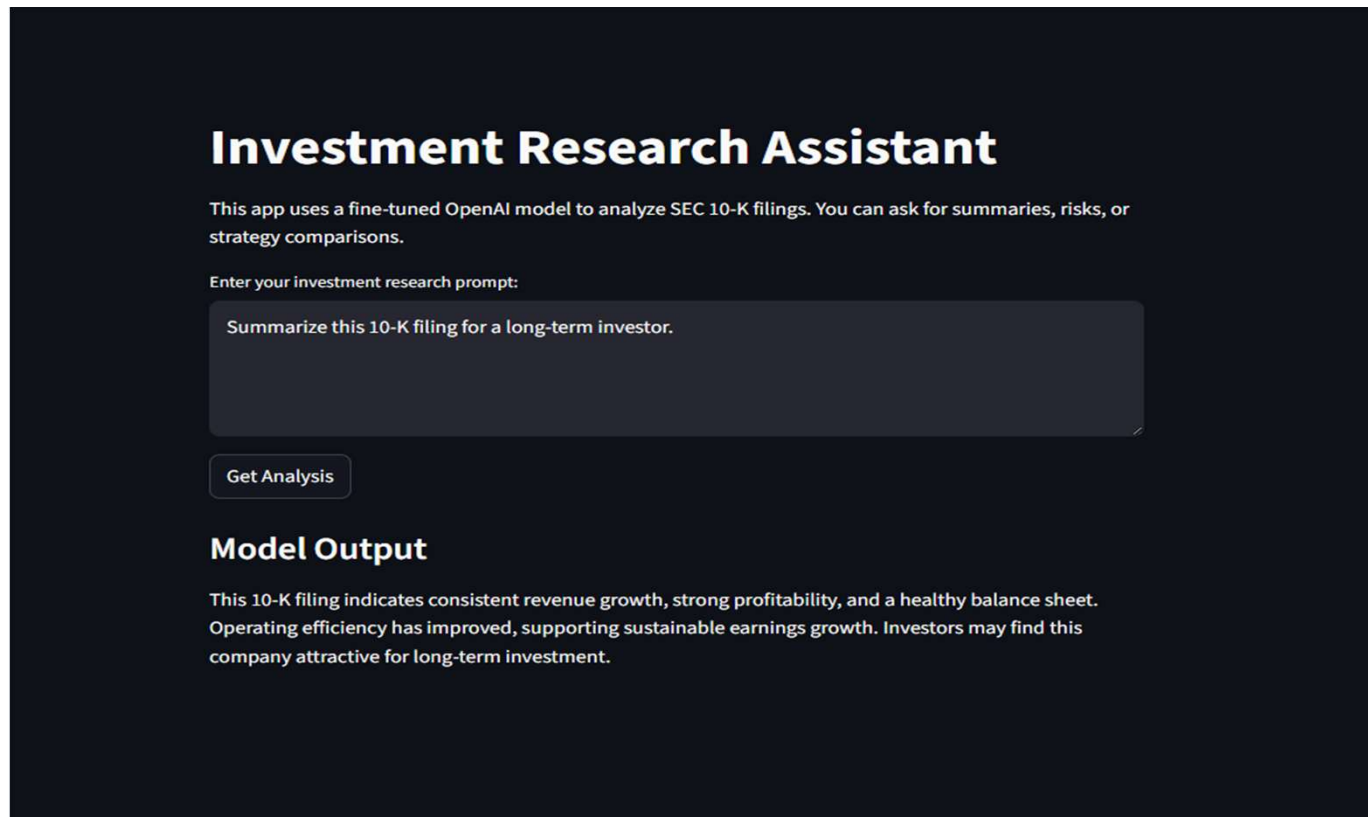
Streamlit Application

- Built an interactive interface using Streamlit
- User enters investment-related prompts
- Application sends a request to the fine-tuned model
- Displays analyst-style response in real-time



Example: Long-Term Investor Summary

- Prompt:
- Summarize this 10-K filing for a long-term investor.
- Model Output:
- This 10-K filing indicates consistent revenue growth, strong profitability, and a healthy balance sheet. Operating efficiency has improved, supporting sustainable earnings growth. Investors may find this company attractive for long-term investment.



Investment Research Assistant

This app uses a fine-tuned OpenAI model to analyze SEC 10-K filings. You can ask for summaries, risks, or strategy comparisons.

Enter your investment research prompt:

Summarize this 10-K filing for a long-term investor.

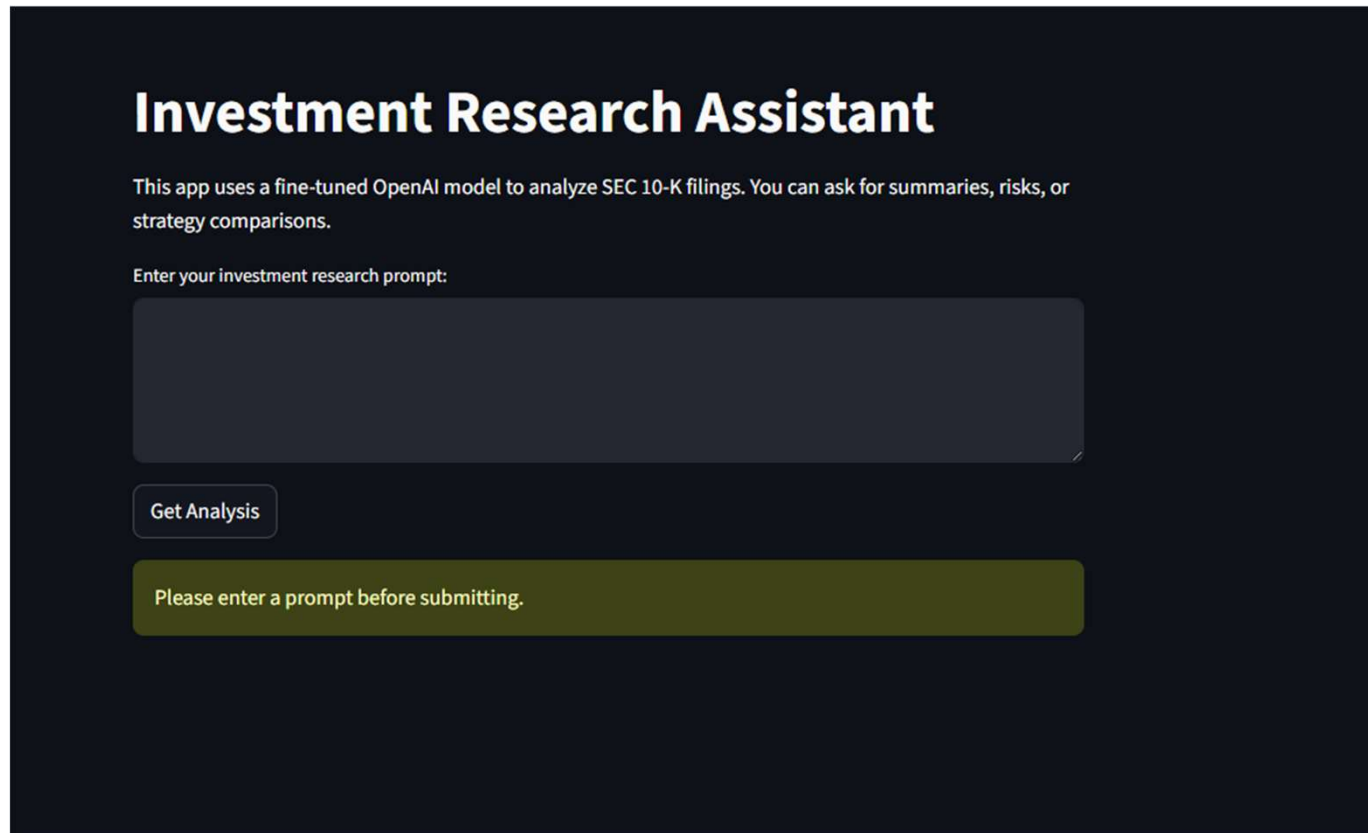
Get Analysis

Model Output

This 10-K filing indicates consistent revenue growth, strong profitability, and a healthy balance sheet. Operating efficiency has improved, supporting sustainable earnings growth. Investors may find this company attractive for long-term investment.

Edge Case Testing

- Tested empty input submission
- Application correctly prevents blank prompts
- Displays a warning message:
- "Please enter a prompt before submitting."



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Key Takeaways

- Successfully fine-tuned a GPT model for financial analysis
- Built a functional Streamlit web application
- Model produces structured investment insights
- Application handles user input validation
- Ready for future expansion using real SEC filings